

Mathematics 2

Homework GD.2

March 26, 2026

The homework consists of six theoretical and practical problems. The solutions are to be submitted as **one .zip file** to the appropriate mailbox on uclnca. The submission should contain a .pdf file containing clear and well described solutions to each of the problems: procedure, proofs and explicit computations (for theoretical problems), code, explanation of choices of parameters, numerical results, etc.

1 Theoretical problems

1. A sequence $\{x_i\}_i$ is eventually p -periodic if for some $N, p \in \mathbb{N}$ we have $x_{N+j} = x_{N+j+p}, \forall j \in \mathbb{N}$. Which of the following procedures on a strictly convex function may result in non-constant 2- or 3-periodic sequences for an appropriate selection of fixed positive parameters γ, μ and starting point x_1 : GD, Polyak GD, Nesterov GD? Explain/prove your answer.
2. Determine the optimal learning rates γ, μ for the Polyak GD for function

$$f(x, y, z) = x^2 + 2y^2 - 2yz + 4z^2 + 3x - 4y + 5z.$$

2 Programming problems

3. Implement GD, Polyak GD, Nesterov GD, and AdaGrad GD.
4. Find, describe, and implement Adam GD.
5. Implement the Newton method and BFGS.
6. Compare the methods of 3., 4, and 5. on:
 - (a) $f(x, y, z) = (x - z)^2 + (2y + z)^2 + (4x - 2y + z)^2 + x + y$ for starting points $(0, 0, 0)$ and $(1, 1, 0)$.
 - (b) $f(x, y, z) = (x-1)^2 + (y-1)^2 + 100(y-x^2)^2 + 100(z-y^2)^2$ for starting points $(1.2, 1.2, 1.2)$ and $(-1, 1.2, 1.2)$.
 - (c) $f(x, y) = (1.5 - x + x y)^2 + (2.25 - x + x y^2)^2 + (2.625 - x + x y^3)^2$ with starting points $(1, 1)$ and $(4.5, 4.5)$.

Describe:

- (a) which one performs best in 2, 5, 10, 100 steps;
 - (b) which one performs best in .1, 1, 2 seconds.
7. Linear regression. Generate N points of the form $\{(i, i + \nu) \mid i = 1, 2, \dots, N\}$, where ν is a random noise on the interval $[0, 1]$. Using the least squares fitting try to fit a linear function $g(x) = kx + n$ to these points. Use GD, SGD, Newton, BFGS and L-BFGS for $N=50, 100, 1000, 10\ 000, 100\ 000, 1\ 000\ 000$ and compare the performances.
 8. Extra problem: Suppose H is a symmetric positive definite matrix, which is also block diagonal. Can you think of an algorithm that will improve the known GD algorithms for functions of the form $f(x) = x^T H x$? If so, describe the solution and justify it as well as possible. Then demonstrate it on an example.